London Ambulance Service NHS
NHS Trust

Clinical Shock

Component 5

Clinical Shock – Component 5

Objective

Explain the types, causes, stages and the management of shock

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Types of Shock

- Hypovolaemic
- Distributive
- Cardiogenic
- Obstructive
- Dissociative

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Hypovolaemic Shock

Acute loss of circulating blood volume

Can be caused by:

- Severe haemorrhage (internal/external) where whole blood volume is lost
- Extensive Burns – Serum is lost
- Severe Diarrhoea +/- Vomiting – Water & electrolytes are lost

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Distributive shock

- Caused by widespread dilation of the peripheral vascular system
- Creates the effect of a larger container for the same blood volume
- In cases of sepsis/anaphylaxis this creates “leaky” blood vessels allowing fluid to move into the interstitial space

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Distributive shock

Common causes of distributive shock

- Anaphylaxis
- Sepsis
- Nervous system related, such as spinal cord injury

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Cardiogenic shock

Causes by the heart being unable to circulate sufficient blood to meet the body's metabolic needs

- Most common following a myocardial infarction
- Acute heart failure
- Arrhythmia

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Obstructive shock

Uncommon cause of shock caused by an obstruction in the flow to/from the heart.

Often caused by

- Tension Pneumothorax
- Cardiac tamponade
- Pulmonary embolism

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Dissociative shock

- Caused by the oxygen carrying capacity of the blood being affected by inadequate number of red blood cells available to carry sufficient oxygen.

- Or when competing molecules take up space on red blood cells

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Dissociative shock

Causes:

- Anaemia
- Carbon monoxide poisoning

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Signs & Symptoms of Shock

- Tachycardia (Increased Pulse Rate)
- Tachypnoea (Increased Respiratory Rate) or Depth
- Abnormal Behaviour (agitation or confusion)
- Feeling dizzy/faint
- Pale , cold extremities, delayed capillary refill
- Sweating, dilated pupils, thirst
- Stage 4 – Near fatal – Reduced LOC, Bradycardia

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Signs & Symptoms of Shock

Certain patient groups may not have abnormal vital signs until significant (1,000ml-1,500ml) has been lost

- Young athletic/healthy people
- Pregnant patients
- Patients on Beta Blocker medication

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Management of shock

- R <c> A <c> B C D E
- Assess for Time Critical
- Administer high % oxygen
- If conscious, position patient flat with head and shoulders slightly raised
- Consider raising lower limbs
- If unconscious, in stable side position
- Treat other injuries and bleeding

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Management of shock

- Do not overheat or allow to become cold
- Avoid rough handling, be gentle
- Constant reassurance
- Constant observations (pulse, level of consciousness, respiration)
- Give nil by mouth – ?need for anaesthetic
- Time Critical transfer to appropriate facility
- Consider paramedic / medical help

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